

Software Engineer Technical Design Round



Company: Orbtronics Ltd

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Challenge Details

Software Developer Technical Design Round

Challenge: “Mini Tasks” — Build & Deploy in 5 Days

Brief

Build a **task manager** (web) where users can:

- Sign up / sign in
- Create/update/delete tasks (title, description, priority, due date, status)
- See tasks in a list with simple filters (status, due this week)
- (Bonus) AI: “Suggest due date” button that proposes a due date from the description

Hard Requirements

- **Frontend:** Next.js + React
- **Backend:** Python
- **DB:** MongoDB
- **Auth:** Email/password (session/JWT) — no social logins
- **Deployment:** Live URL(s) required
- **API:** REST JSON
- **Testing:** At least 3 minimal tests (unit or API)
- **Docs:** README.md with setup, env vars, deployment notes, and architecture diagram (can be a PNG or Mermaid)

Nice to Have (Bonus)

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- AI due date suggestion via **OpenAI API** (gpt4omini or similar)
- Realtime task updates (WebSocket/SSE) or optimistic UI
- Basic responsive design & accessibility checks
- Dockerfiles for both services

Deliverables

- Public Git repo with:
 - frontend/
 - backend/
 - README.md (how to run locally & URLs to deployed app + API)
 - Architecture diagram (Mermaid or image)
 - Short note on tradeoffs & future improvements
- Live URLs:
 - Web app URL
 - API base URL
- Demo credentials (a nonadmin test user)

Timebox

- **Take home window:** 5 days from receipt of Technical Design Round Email

Evaluation Rubric (100 pts)

- **Correctness & Requirements (25)** — Auth works, CRUD works, filters work, deployed
- **Code Quality (20)** — Clear structure, readable, types where sensible, linting

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- **API & Data Design (15)** — Sensible models, status codes, validation
- **Security & Secrets (10)** — Hashing (bcrypt/argon2), JWT/session safety, CORS, env vars
- **DX & Docs (10)** — Clean README, run scripts, .env.example, diagram
- **Testing (10)** — Basic tests pass locally/CI
- **Deployment (5)** — Stable DO deployment, health checks
- **Bonus Features (up to 5)** — OpenAI duedate, realtime, Docker

Guardrails

- May use UI libs (e.g., Tailwind, shadcn/ui) but **no full CRUD templates/starters**.
- Must write your own auth flow; ok to use a small JWT lib.
- If using the OpenAI bonus, **explain prompt + safeguards** and keep the key serverside.

Scoring Guide (what we look for)

- **L1 pass:** Solid CRUD + auth, clean repo, deployed, basic tests, minimal bugs
- **Strong:** Good architecture notes, thoughtful validation, nice UX polish, small bonus complete
- **Exceptional:** Bonus + Docker, CI, secure headers, clear tradeoff notes